Ruijie Zheng

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Education

Shanghai Jiao Tong University (SJTU), Joint Institute — Sep.2023 – July.2025
Undergraduate student in Mechanic Engineering (ME) — GPA: 3.78/4.0 (48/335)

University of Michigan, Ann Arbor, Engineering college — Aug.2025 – Present
Undergraduate student in Computer Science (CS) — GPA: 3.933/4.0

Project and Research Experience

Course Project: LLM-Driven Tennis Coaching System — Sep.2025 – Dec.2025
University of Michigan, Ann Arbor Computer Vision

- Developed an end-to-end automated pipeline extracting pose landmarks from tennis videos using MediaPipe and Sport2D
- Designed a heuristic keyframe selection algorithm and a phase-aware LLM prompting strategy to synthesize personalized natural-language coaching feedback

Research Project: Sparse Weight-Decomposed LoRA (SDoRA) — Aug.2025 – Dec.2025
University of Michigan, Ann Arbor

- Proposed SDoRA, a unified Parameter-Efficient Fine-Tuning (PEFT) framework combining weight decomposition with learnable sparse gating
- Implemented the SoRA module and trained Qwen2.5-3B models to analyze sparsity dynamics and layer-wise pruning patterns
- Conducted extensive experiments on commonsense reasoning benchmarks (BoolQ, PIQA, ARC-c) to evaluate training stability and performance

Research Program: Water Jet Propulsion Based on Negative Pressure Energy Storage
Nov.2024 – Present
School of Mechanical Engineering, SJTU

- Construct the algorithm logic using Simulink and optimize the control through reinforcement learning
- Modeling the spacecraft shell using SolidWorks

Innovation Program: Cross twin rotor unmanned helicopter — Feb.2024 – Oct.2024
School of Electronic Information and Electrical Engineering, SJTU

- Used quaternions to implement the control algorithm of the aircraft
- Assembled part of the aircraft

Research Program: Multi-modality Federated Learning — Jan.2024 – Oct.2024
School of Electronic Information and Electrical Engineering, SJTU

- Used pytorch to reproduce some algorithms in federated learning, such as Fedavg, Harmony, etc
- Improved the efficiency of the algorithm in processing multi-modal data

Course Design: Mars rover model — May.2024 – Aug.2024
Joint Institute, SJTU

- Took ENGR1000J Intro to Engineering and was responsible for the 3D modeling of the Mars rover
- Assembled the car and implemented part of the control algorithm of the rover

Skills

- **Software**: MATLAB, SolidWorks, Mathematica, Powerpoint
- **Design & Prototyping**: Arduino, Raspberry pi, 3D printing, CO2 laser cutting

Leadership

Shanghai Jiao Tong University, Joint Institute — Sep.2024 – Present

The Vice President of American Society of Mechanical Engineers (ASME) student branch

- Held workshops and lectures on mechanical engineering topics for students in Joint Institute
- Maintain the ASME official webpage in Joint Institute

Shanghai Jiao Tong University, Joint Institute — Sep.2024 – Jan.2025

Teaching Assistant for MATH1860J Honors Mathematics

- Held recitation class and office hours, prepared worksheets for students to review
- Graded homework and exams, held review sessions before exams

Shanghai Jiao Tong University — May.2024 and Aug.2024

Volunteer teaching activity Yingshang, Anhui

- Deliver lectures on basic physics and mathematics for primary school students in rural areas
- Organize extracurricular activities for students